Code No: 136CF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
INTRODUCTION TO ANALYTICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Used paper Aug-2024

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is structured array? Give example.	[2]
b)	What are the challenges in analysis of data?	[3]
c)	Define sample space.	[2]
d)	Calculate clear statistics for random variables.	[3]
e)	What is R connector?	[2]
f)	What are the benefits of NOSQL databases?	[3]
g)	How correlation is used for data analysis?	[2]
h)	What is regression? Explain it briefly.	[3]
i)	What is business intelligence?	[2]
j)	What are problems related to a business?	[3]

PART – B

(50 Marks)

2.a) Explain data sets in R with their syntax and example.
b) Discuss the R studio in detail. [5+5]

OR

3.a) How do you find outliers in R explain with an example?
b) Write a R program for matrix multiplication. [5+5]

4.a) What is summarization of data? Explain different methods of summarizations in R.
b) Elaborate on population and theoretical distribution. [5+5]

OR

5.a) What is probability function. Calculate probability for n*(n-1) functions.
b) Discuss the various forms of nonverbal communication in data analytics in detail. [5+5]

6.a) Explain the role of NoSQL in data analytics using an example.
b) Write about stored procedures in NOSQL. Explain with examples. [5+5]

OR

7.a) Differentiate between data integrity and data manipulator.
b) How database connector is used to R-integrity? [5+5]

8.a) What is the relationship between covariance and correlation coefficient. How they can be computed in R?

b) Give the importance of correlation for prediction. [5+5]

OR

9.a) Explain about Heteroscedasticity and auto correlation.

b) Enumerate the types of regression. Explain them with example. [5+5]

10.a) Explain about automated mechanism design.

b) Write an optimality function for a close loop system. [5+5]

OR

11.a) Compare and contrast the manufacturing and production activities.

b) Explain in detail about SMART utilities. [5+5]

---ooOoo---

Used paper Aug-2024